

ACADIA VALLEY, AB, Canada

WMO: 710480

Lat: 51.066N

Lon: 110.317W

Elev: 735

StdP: 92.80

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.4	-26.7	-32.8	0.2	-29.2	-29.8	0.3	-26.6	13.9	-2.8	12.3	-0.4	2.9	270	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	31.2	16.6	29.2	16.2	27.3	15.8	18.9	26.0	17.9	25.1	17.1	24.5	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.9	21.6	15.4	12.0	20.7	14.4	11.2	19.5	56.8	25.8	53.5	25.1	50.8	24.8	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.8	11.0	9.5	DB	-33.8	34.5	1.7	1.8	-35.1	35.8	-36.1	36.9	-37.1	37.9	-38.3	39.2
			WB	-33.9	21.4	1.7	1.2	-35.2	22.3	-36.2	23.0	-37.2	23.7	-38.4	24.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-9.6	-11.1	-3.3	4.7	11.4	15.7	18.9	18.1	13.1	5.2	-3.6	-10.5		
	DBStd	12.47	8.18	8.14	8.15	5.11	4.50	3.29	2.83	3.22	4.93	5.30	7.48	7.94		
	HDD10.0	3050	607	589	412	169	37	3	0	1	25	165	408	634		
	HDD18.3	5273	865	823	669	409	216	91	28	46	167	409	657	893		
	CDD10.0	923	0	0	0	9	82	173	274	251	118	14	1	0		
	CDD18.3	106	0	0	0	0	3	12	44	37	11	0	0	0		
	CDH23.3	1694	0	0	0	4	109	197	606	549	223	5	0	0		
	CDH26.7	527	0	0	0	0	21	49	197	182	78	0	0	0		
(14) Wind	WSAvg	4.4	4.7	4.3	4.5	5.0	4.7	4.5	3.9	3.8	4.1	4.7	4.4	4.3		
(15) Precipitation	PrecAvg	330	14	8	13	21	37	73	53	43	25	14	13	11		
	PrecMax	471	47	18	36	61	80	150	128	84	83	33	44	18		
	PrecMin	172	2	1	3	3	3	17	20	5	1	4	0	2		
	PrecStd	75	10	5	7	15	23	32	24	21	21	7	11	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	7.7	16.9	23.1	29.3	31.0	33.8	33.1	32.3	23.8	17.6	8.2		
		MCWB	4.1	3.7	8.6	10.5	14.2	16.4	18.4	16.4	15.6	12.6	8.9	4.1		
	2%	DB	5.4	4.4	13.5	20.1	26.3	27.7	31.0	30.8	29.3	19.6	12.0	5.2		
		MCWB	2.6	1.6	6.5	9.3	13.1	15.8	17.5	16.2	14.8	10.5	5.9	2.1		
	5%	DB	3.5	2.4	10.2	17.0	23.9	25.4	28.7	28.8	25.8	16.6	8.4	2.6		
		MCWB	1.3	0.5	4.8	7.5	12.0	15.4	17.3	16.0	13.9	9.1	3.9	0.2		
	10%	DB	1.8	0.5	6.9	14.1	21.0	23.5	26.8	26.6	22.7	13.6	5.5	0.2		
		MCWB	0.1	-0.8	3.4	6.1	10.8	14.1	16.8	15.7	12.6	7.5	2.5	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	4.2	8.9	11.0	14.9	19.1	21.1	19.3	16.7	13.4	9.4	4.6		
		MCDB	8.0	7.4	16.1	22.0	26.1	25.8	29.9	26.8	28.7	22.0	17.3	7.6		
	2%	WB	2.7	1.9	7.0	9.4	13.7	17.5	19.5	18.1	15.4	11.4	6.5	2.2		
		MCDB	5.2	4.0	12.5	18.9	24.1	24.6	26.6	25.8	26.9	17.8	11.4	5.1		
	5%	WB	1.5	0.7	5.2	8.0	12.7	16.2	18.5	17.3	14.4	9.5	4.3	0.5		
		MCDB	3.3	2.3	10.0	15.7	21.8	23.2	25.5	25.3	24.2	15.4	7.7	2.5		
	10%	WB	0.1	-0.9	3.2	6.6	11.5	15.2	17.6	16.4	13.1	8.0	2.8	-1.4		
		MCDB	1.8	0.5	6.6	13.2	19.3	21.6	24.5	24.3	22.0	13.5	5.1	0.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	9.9	10.6	12.6	13.6	12.4	13.7	13.7	13.8	12.2	10.1	9.0		
		MCDBR	10.2	11.2	15.2	18.3	17.6	15.8	17.5	18.2	19.6	16.8	14.5	11.4		
	5% WB	MCWBR	8.0	8.7	9.0	9.2	7.6	6.8	6.7	6.5	7.9	8.8	9.3	8.8		
		MCDWR	10.2	11.0	14.5	16.8	15.5	13.9	14.7	15.0	18.2	15.8	13.3	11.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.263	0.282	0.295	0.350	0.377	0.367	0.371	0.388	0.332	0.294	0.264	0.253		
		taud	2.367	2.376	2.444	2.298	2.265	2.356	2.370	2.285	2.428	2.509	2.477	2.383		
	Ebn at Noon	Ebn at Noon	778	859	915	889	877	888	878	840	859	838	781	734		
		Edh at Noon	61	79	90	117	127	117	114	118	91	69	54	51		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.18	2.21	3.58	4.88	5.86	6.14	6.63	5.38	3.87	2.35	1.26	0.89		
	RadStd	RadStd	0.12	0.20	0.26	0.37	0.39	0.39	0.30	0.38	0.34	0.28	0.11	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page